



CIE A Level Chemistry



Your notes

5.1 Enthalpy Change, ΔH

Contents

- * Enthalpy Change, ΔH
- * Reaction Pathway Diagrams
- * Standard Enthalpy Change
- * Enthalpy & Bond Energies



Your notes

Enthalpy Change, ΔH

Enthalpy Changes

- The total chemical energy inside a substance is called the enthalpy (or heat content)
- When chemical reactions take place, changes in chemical energy take place and therefore the enthalpy changes
- An enthalpy change is represented by the symbol ΔH (Δ = change; H = enthalpy)
- An enthalpy change can be positive or negative

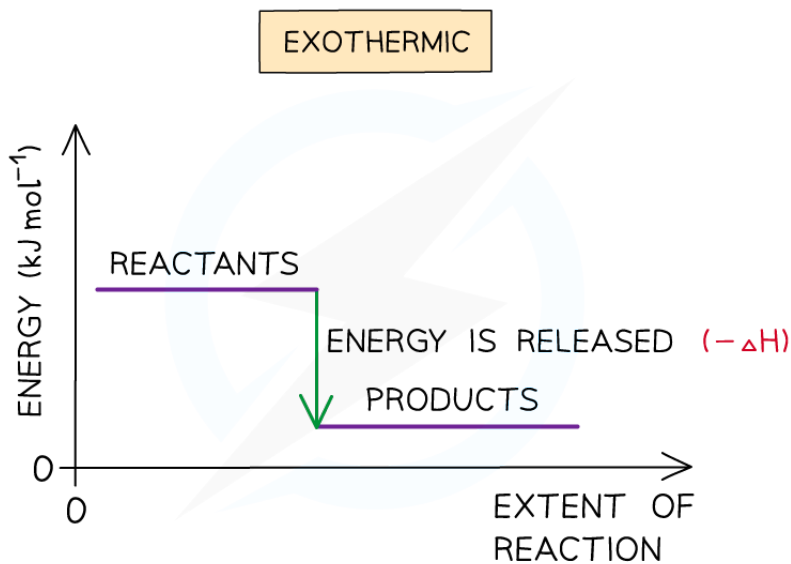
Exothermic reactions

- A reaction is exothermic when the products have less energy than the reactants
- Heat energy is **given off** by the reaction **to the surroundings**
 - The **temperature** of the **environment increases** - this can be measured on a thermometer
 - The **temperature** of the **system decreases**
- There is an enthalpy decrease during the reaction so ΔH is negative
- Exothermic reactions are **thermodynamically** possible (because the enthalpy of the reactants is **higher** than that of the products)
- However, the rate may be too slow to observe any appreciable reaction
 - In this case the reaction is **kinetically** controlled
 - This means the reaction could have a high activation energy which is preventing the reaction from taking place.

The enthalpy change during an exothermic reaction



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Exothermic reactions show a negative enthalpy change as energy is released

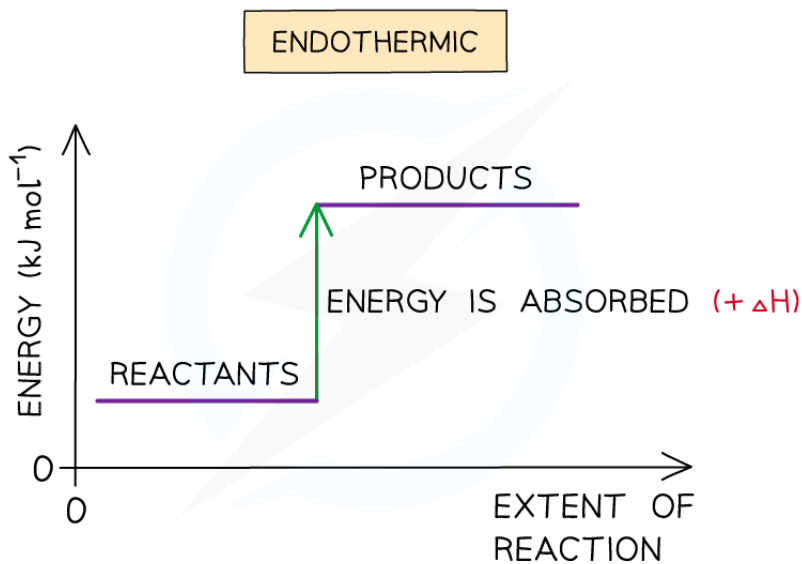
Endothermic reactions

- A reaction is endothermic when the products have more energy than the reactants
- Heat energy is **absorbed by** the reaction **from** the **surroundings**
 - The **temperature** of the **environment decreases** - this can be measured with a thermometer
 - The **temperature** of the **system increases**
- There is an enthalpy increase during the reaction so ΔH is positive

The enthalpy change during an endothermic reaction



Your notes



Endothermic reactions show a positive enthalpy change as energy is absorbed

Exam Tip

- It is important to specify the physical states of each species in an equation when dealing with enthalpy changes as any changes in state can cause very large changes of enthalpy.
- For example:



- Also, remember that the **system** is the **molecules that are reacting** (i.e. the reaction itself) and the **surroundings are everything else** (eg. the flask the reaction is taking place in).



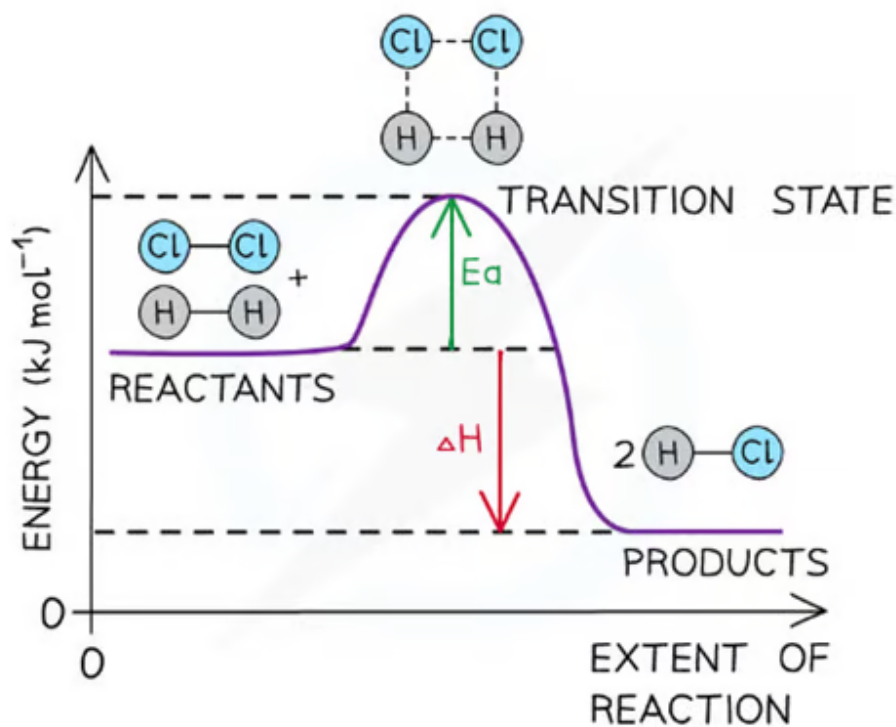
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Reaction Pathway Diagrams

Reaction Pathway Diagrams

- A reaction pathway diagram shows the energies of the reactants, the transition state(s) and the products of the reaction with time
 - They are sometimes called energy profile diagrams
- The **transition state** is a stage during the reaction at which chemical bonds are partially broken and formed
- The transition state is very unstable – it cannot be isolated and is higher in energy than the reactants and products
- The activation energy (E_a) is the energy needed to reach the transition state
- We can define the activation energy as '*the minimum amount of energy needed for reactant molecules to have a successful collision and start the reaction*'

Example reaction pathway diagram



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The reaction pathway diagram for the reaction of hydrogen with chlorine to form hydrogen chloride gas

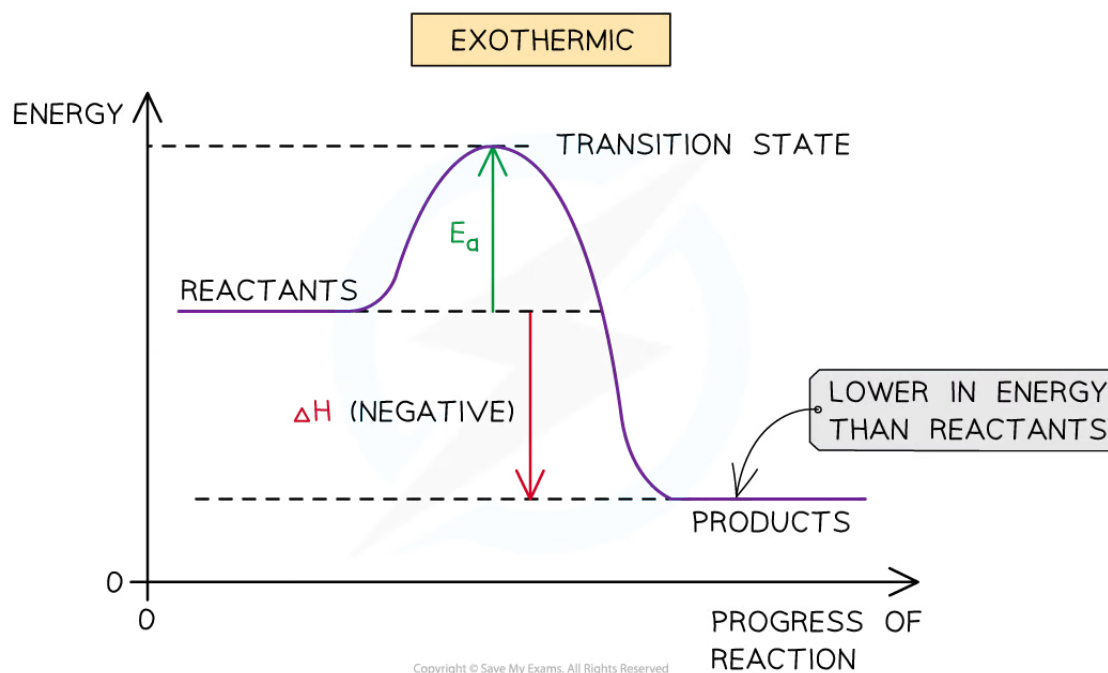


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Exothermic reaction

- In an exothermic reaction, the reactants are higher in energy than the products
- The reactants are therefore closer in energy to the transition state
- This means that exothermic reactions have a lower activation energy compared to endothermic reactions

The reaction pathway diagram for exothermic reactions



Exothermic reaction pathway diagrams always have the products lower in energy than the reactants

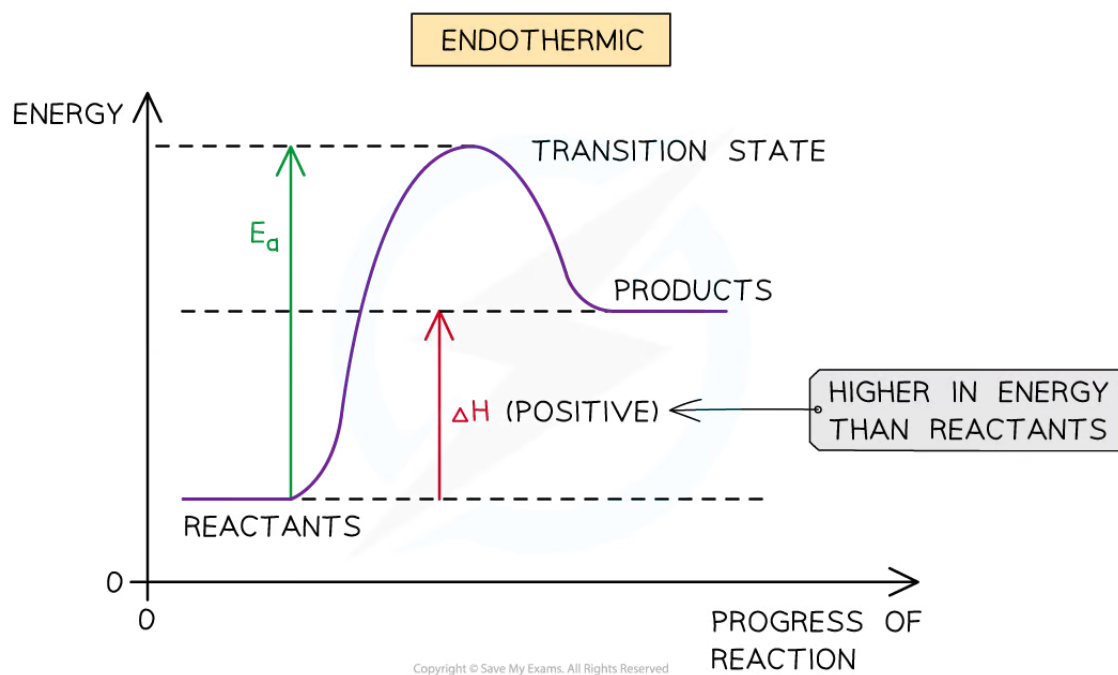
Endothermic reaction

- In an endothermic reaction, the reactants are lower in energy than the products
- The reactants are therefore further away in energy to the transition state
- This means that endothermic reactions have a higher activation energy compared to exothermic reactions

The reaction pathway diagram for endothermic reactions



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Endothermic reaction pathway diagrams always have the products higher in energy than the reactants



Your notes

Worked example

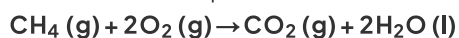
Drawing energy level diagrams of the combustion of methane

The E_a and ΔH for the complete combustion of methane are $+2653 \text{ kJ mol}^{-1}$ and -890 kJ mol^{-1} respectively.

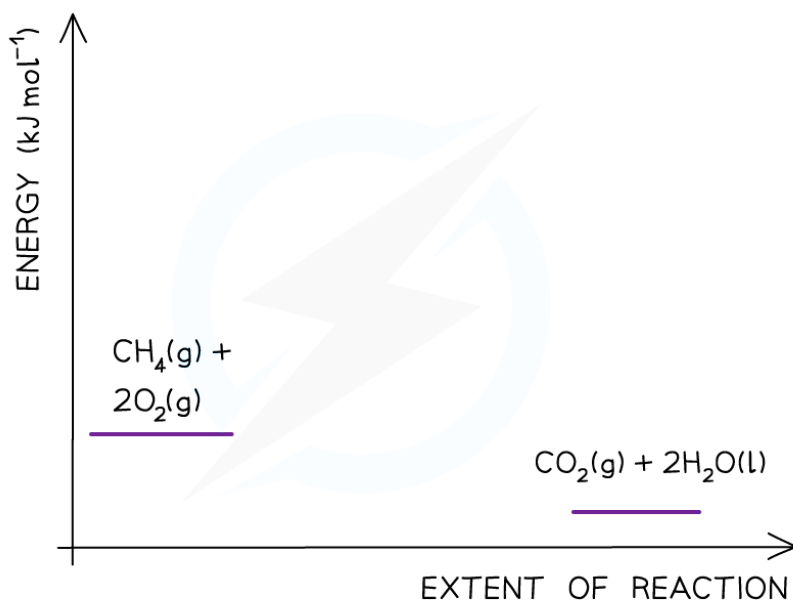
Draw the reaction pathway diagram for this reaction

Answer

- Step 1:** The chemical equation for the complete combustion of methane is:



- Step 2:** Combustion reactions are always exothermic (ΔH is negative) so the reactants should be drawn higher in energy than the products

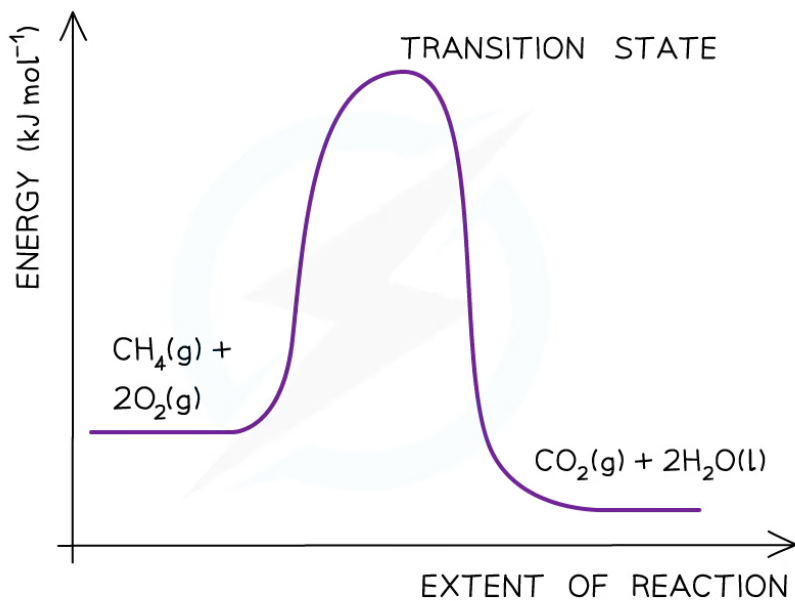


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- Step 3:** Draw the curve in the energy level diagram clearly showing the transition state

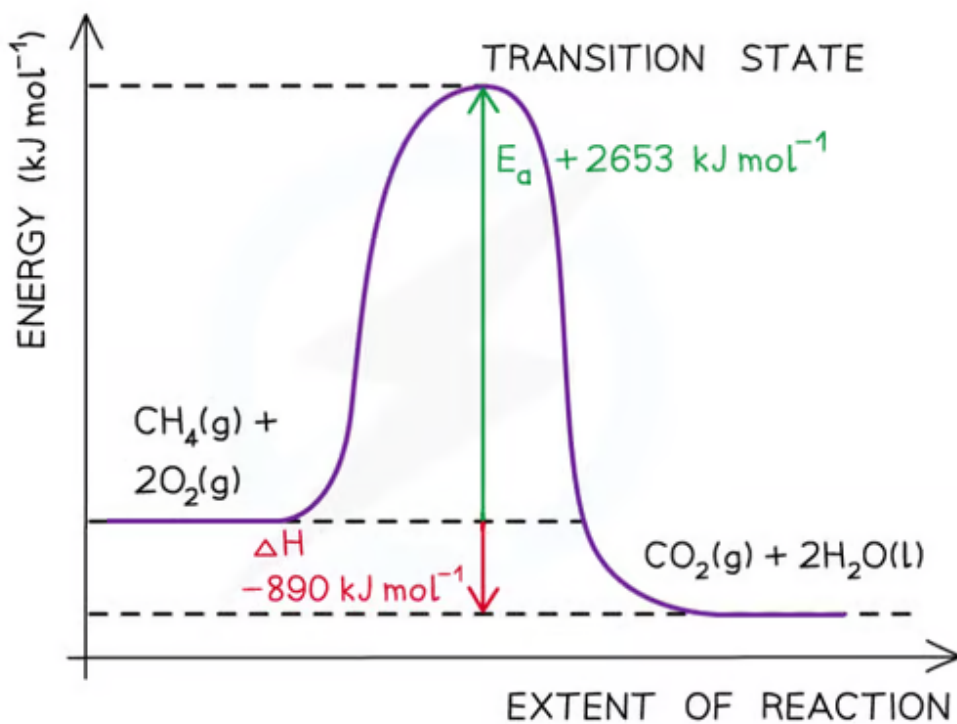


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- **Step 4:** Draw arrows to show the E_a and ΔH including their values



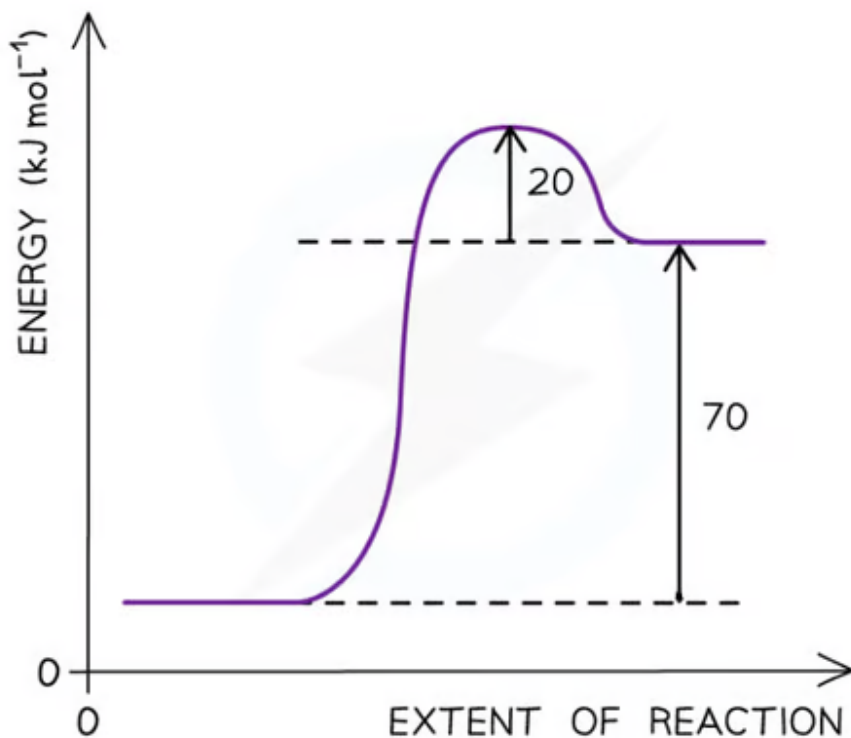


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Worked example

Determining the activation energy

Determine, using the reaction pathway diagram, the E_a for the forward and backward reaction.



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Answer

- The E_a is the energy difference from the energy level of the reactants to the top of the 'hump'
- $E_a(\text{forward reaction}) = (+70 \text{ kJ mol}^{-1}) + (+20 \text{ kJ mol}^{-1}) = +90 \text{ kJ mol}^{-1}$
- As the question is asking for the **reverse reaction** the E_a is the energy difference from the energy level of the products to the 'hump'
- $E_a(\text{reverse reaction}) = +20 \text{ kJ mol}^{-1}$

 Exam Tip

- The activation energy is the energy difference from **reactants** to the **transition state**.
- The enthalpy change of the reaction is the energy difference from **reactants** to **products**.
- Remember to label the axis of the reaction pathway diagrams!



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Standard Enthalpy Change

Defining Enthalpy Changes

- To fairly compare the changes in enthalpy between reactions, all reactions should be carried out under standard conditions
- These standard conditions are:
 - A **pressure** of 101 kPa
 - A **temperature** of 298 K (25 °C)
 - Each substance involved in the reaction is in its normal physical state (solid, gas or liquid)
- To show that a reaction has been carried out under standard conditions, the symbol θ is used
 - ΔH^θ = the standard enthalpy change
- These are a number of key definitions for common language relating to enthalpy change that all chemists need to know

Enthalpy definitions table

Standard Enthalpy Change of...	Definition	Symbol	Exothermic / Endothermic
Reaction	The enthalpy change when the reactants in the stoichiometric equation react to form the products under standard conditions	ΔH^θ_r	Both
Formation	The enthalpy change when one mole of a compound is formed from its elements under standard conditions	ΔH^θ_f	Both
Combustion	The enthalpy change when one mole of a substance is burnt in excess oxygen under standard conditions	ΔH^θ_c	Exothermic
Neutralisation	The enthalpy change when one mole of water is formed by reacting an acid and an alkali under standard conditions	ΔH^θ_{neut}	Exothermic



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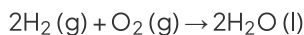
Worked example

Calculating the enthalpy change of reaction of water

One mole of water is formed from hydrogen and oxygen, releasing 286 kJ of energy.



Calculate $\Delta H_f^\ominus$ for the reaction below:



Answer

- The $\Delta H_f^\ominus$ value of -286 kJ mol^{-1} is for one mole of water being formed
- Two moles of water molecules are formed for the equation in question
- So, the energy released is simply:
 - $\Delta H_f = 2 \text{ mol} \times (-286 \text{ kJ mol}^{-1})$
 - $\Delta H_f = -572 \text{ kJ mol}^{-1}$

Worked example

Calculating the enthalpy change of formation

Calculate $\Delta H_f^\ominus$ of the reaction below:



Answer

- The $\Delta H_f^\ominus$ value of $-824.2 \text{ kJ mol}^{-1}$ is for one mole of $\text{Fe}_2\text{O}_3(\text{s})$ being formed
- Two moles of $\text{Fe}_2\text{O}_3(\text{s})$ are formed for the equation in question
- So, the energy released is simply:
 - $\Delta H_f^\ominus = 2 \text{ mol} \times (-824.2 \text{ kJ mol}^{-1})$
 - $\Delta H_f^\ominus = -1648.4 \text{ kJ mol}^{-1}$



Your notes

Worked example

Calculating enthalpy changes

Identify each of the following as ΔH^θ_r , ΔH^θ_f , ΔH^θ_c or ΔH^θ_{neut} :

- $\text{MgCO}_3(\text{s}) \rightarrow \text{MgO}(\text{s}) + \text{CO}_2(\text{g})$
- $\text{C}(\text{graphite}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$
- $\text{HCl}(\text{aq}) + \text{NaOH}(\text{aq}) \rightarrow \text{NaCl}(\text{aq}) + \text{H}_2\text{O}(\text{l})$

Answers:

Answer 1

- It cannot be ΔH^θ_r as there is more than one compound being formed
- It cannot be ΔH^θ_c as there is no reaction with oxygen
- It cannot be ΔH^θ_{neut} as there is no acid or alkali involved
- This is just a chemical reaction, therefore, it is ΔH^θ_r

Answer 2

- ΔH^θ_f as one mole of CO_2 is formed from its elements in their standard states

AND

ΔH^θ_c as one mole of carbon is burnt completely in oxygen

Answer 3

- ΔH^θ_{neut} as one mole of water is formed from the reaction of an acid and alkali

Exam Tip

The ΔH^θ_f of an **element** in its standard state is zero. For example, ΔH^θ_f of $\text{O}_2(\text{g})$ is 0 kJ mol^{-1}



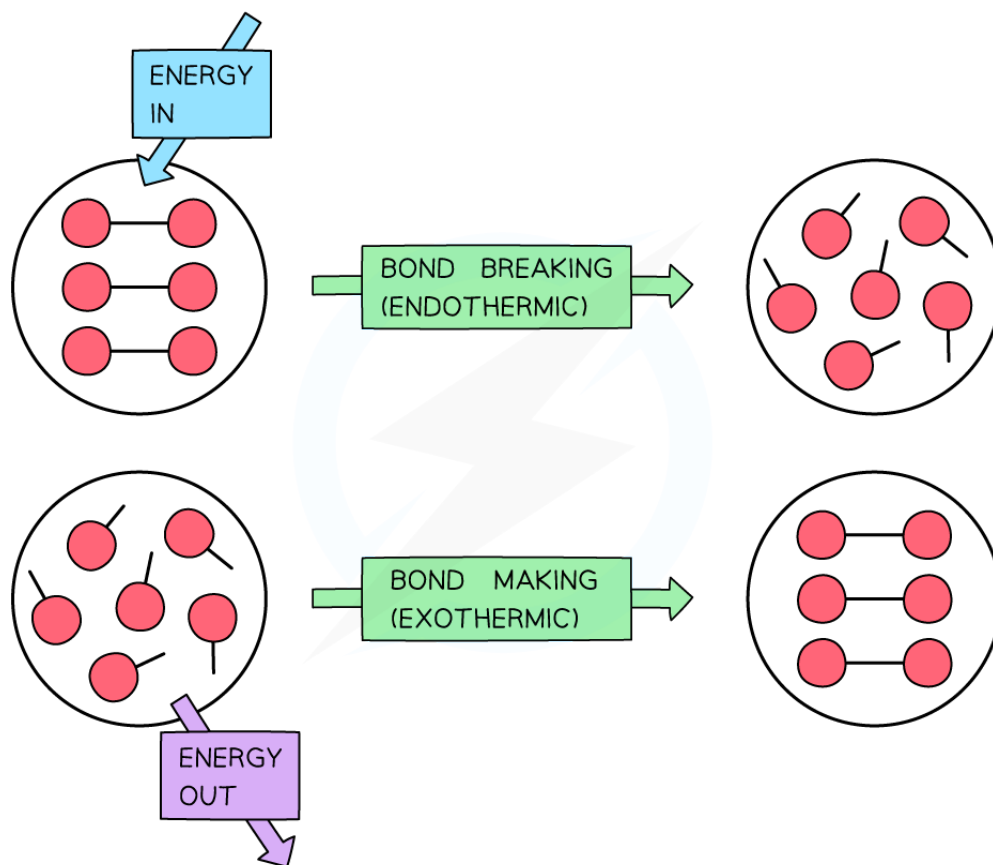
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Enthalpy & Bond Energies

Enthalpy & Bond Energies

- During a reaction, enthalpy changes take place because bonds are being broken and formed
- Energy (in the form of heat) is needed to overcome attractive forces between atoms
- Bond breaking** is therefore **endothermic**
- Energy is released from the reaction to the surroundings (in the form of heat) when new bonds are formed
- Bond forming** is therefore **exothermic**

Making and breaking bonds



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Breaking bonds requires energy from the surroundings and making bonds releases energy to the surroundings

- If more energy is required to break bonds than energy is released when new bonds are formed, the reaction is **endothermic**
- If more energy is released when new bonds are formed than energy is required to break bonds, the reaction is **exothermic**
- In reality, only some bonds in the reactants are broken and then new ones are formed



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Enthalpy Calculations

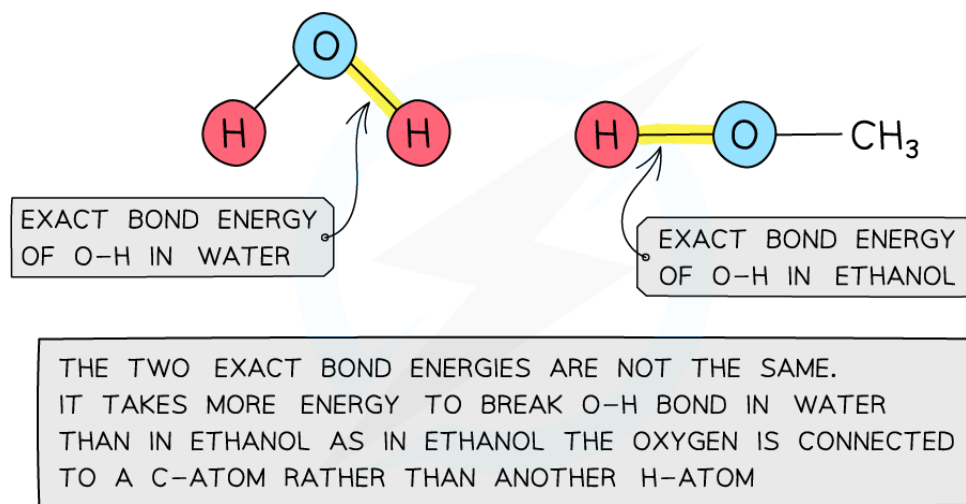
Exact bond energy

- The amount of energy required to break one mole of a **specific** covalent bond in the gas phase is called the bond dissociation energy
- Bond dissociation energy (E) is also known as **exact bond energy** or *bond enthalpy*
- The type of bond broken is put in brackets after E
 - Eg. $E(\text{H-H})$ is the bond energy of a mole of single bonds between two hydrogen atoms

Average bond energy

- Bond energies are affected by other atoms in the molecule (the environment)
- Therefore, an average of a number of the **same type of bond** but in different environments is calculated
- This bond energy is known as the **average bond energy**
- Since bond energies cannot be determined directly, **enthalpy cycles** are used to calculate the average bond energy

How bond energies are affected by other atoms in the molecule



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Bond energies are affected by other atoms in the molecule

Calculating enthalpy change from bond energies

- Bond energies are used to find the $\Delta H_r^\ominus$ of a reaction when this cannot be done experimentally
 - Eg. the Haber Process
- The equation to calculate the standard enthalpy change of reaction using bond energies is:

$$\Delta H_r^\theta = \text{enthalpy change for bonds broken} + \text{enthalpy change for bonds formed}$$



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Worked example

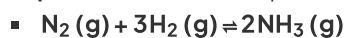
Calculate the enthalpy change of reaction the Haber process reaction.

The relevant bond energies are given in the table.

Bond	Average bond energy kJ mol ⁻¹
N $\equiv$ N	945
H-H	436
N-N	391

Answer

- **Step 1:** The chemical equation for the Haber process is:



- **Step 2:** Set out the calculation as a balance sheet:

Bonds broken kJ mol ⁻¹	Bonds formed kJ mol ⁻¹
1 x N $\equiv$ N = 1 x 945 = 945	6 x N-H = 6 x 391 = 2346
3 x H-H = 3 x 436 = 1308	
Total = +2253	Total = -2346

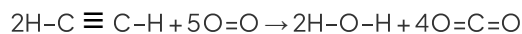
- **Note:** Values for bonds broken are positive (endothermic) and values for bonds formed are negative (exothermic)
- **Step 3:** Calculate the standard enthalpy of reaction:
 - $\Delta H_r^\theta = \text{enthalpy change for bonds broken} + \text{enthalpy change for bonds formed}$
 - $\Delta H_r^\theta = (+2253 \text{ kJ mol}^{-1}) + (-2346 \text{ kJ mol}^{-1})$
 - $\Delta H_r^\theta = -93 \text{ kJ mol}^{-1}$



Your notes

Worked example

The complete combustion of ethyne, C_2H_2 , is shown in the equation below:



Calculate, using the average bond enthalpies given in the table, the enthalpy of combustion of ethyne.

Bond	Average bond energy kJ mol^{-1}
C-H	414
$C \equiv C$	839
O=O	498
C=O	804
O-H	463
O-C	358

Answer

Step 1:

- The enthalpy of combustion is the enthalpy change when **one mole** of a substance reacts in excess oxygen to produce water and carbon dioxide
- The chemical reaction should be therefore simplified such that only **one mole** of **ethyne** reacts in excess oxygen:



- Step 2:** Set out the calculation as a balance sheet:

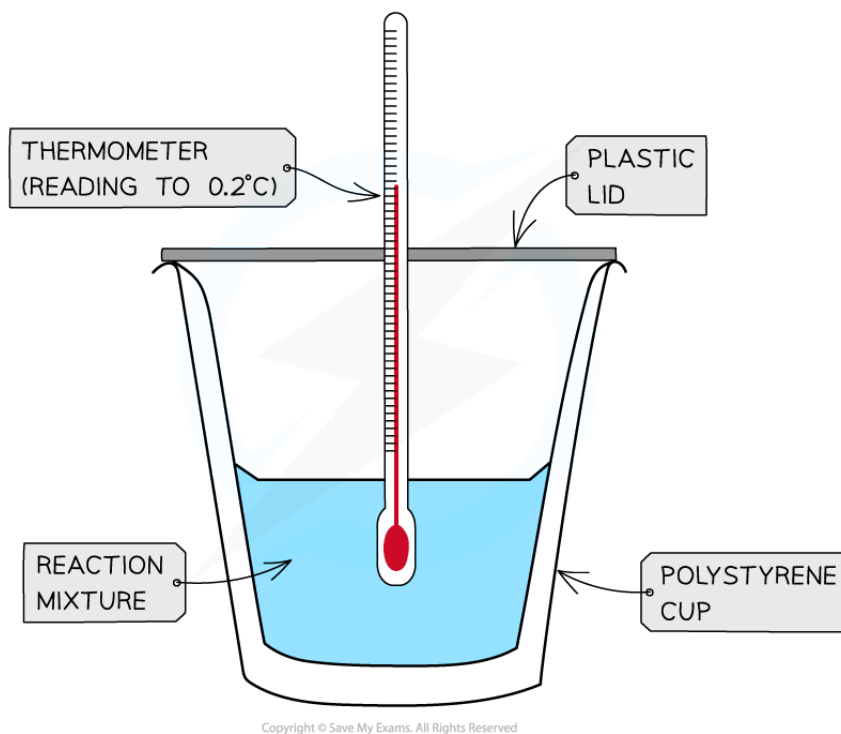
Bonds broken kJ mol^{-1}	Bonds formed kJ mol^{-1}
$1 \times C \equiv C = 1 \times 839 = 839$	$2 \times O-H = 2 \times 463 = 926$
$2 \times C-H = 2 \times 414 = 828$	$4 \times C=O = 4 \times 804 = 3216$

$2\frac{1}{2} \times \text{O}=\text{O} = 2\frac{1}{2} \times 498 = 1245$	
Total = +2912	Total = -4142

- **Note:** Values for bonds broken are positive (endothermic) and values for bonds formed are negative (exothermic)
- **Step 3:** Calculate the standard enthalpy of reaction:
 - ΔH_r^θ = enthalpy change for bonds broken + enthalpy change for bonds formed
 - $\Delta H_r^\theta = (+2912 \text{ kJ mol}^{-1}) + (-4142 \text{ kJ mol}^{-1})$
 - $\Delta H_r^\theta = -1230 \text{ kJ mol}^{-1}$

Measuring enthalpy changes

- **Calorimetry** is a technique used to measure changes in enthalpy of chemical reactions
 - A **calorimeter** can be made up of a **polystyrene drinking cup**, a **vacuum flask** or **metal can**
- Example calorimetry equipment**



A polystyrene cup can act as a calorimeter to find enthalpy changes in a chemical reaction

- The energy needed to increase the temperature of 1g of a substance by 1°C is called the **specific heat capacity** (c) of the liquid
- The specific heat capacity of water is $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$
- The energy transferred as heat can be calculated using the following equation:

$$q = m \times c \times \Delta T$$

- q = the heat transferred, J
- m = the mass of water, g
- c = the specific heat capacity, $\text{J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$ or $\text{J g}^{-1} \text{ K}^{-1}$
- T = the temperature change, $^{\circ}\text{C}$



Your notes



Your notes

Worked example

In a calorimetric experiment, 2.50 g of methane is burnt in excess oxygen.

30% of the energy released during the combustion is absorbed by 500 g of water, causing the temperature to rise from 25 °C to 68 °C.

The specific heat capacity of water is 4.18 J g⁻¹ K⁻¹.

Calculate the total energy released per gram of methane burnt.

Answer

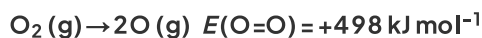
- **Step 1:** Gather the necessary values for the $q = mc\Delta T$ equation:
 - $m(\text{of water}) = 500 \text{ g}$
 - $c(\text{of water}) = 4.18 \text{ J g}^{-1} \text{ K}^{-1}$
 - $\Delta T(\text{of water}) = 68 \text{ }^{\circ}\text{C} - 25 \text{ }^{\circ}\text{C} = 43 \text{ }^{\circ}\text{C} = 43 \text{ K}$
 - The change in temperature in °C is equal to the change in temperature in K
- **Step 2:** Complete the calculation:
 - $q = 500 \times 4.18 \times 43$
 - $q = 89\,870 \text{ J}$
- **Step 3:** Conversion from 30% to 100%
 - The value of q calculated is only 30% of the total energy released by 2.50 g of methane
 - Total energy $\times 0.3 = 89\,870 \text{ J}$
 - Total energy = 299 567 J
- **Step 4:** Calculate the energy released by 1.00 g of methane
 - The energy calculated is released by 2.50 g of methane
 - Energy released by 1.00 g of methane = $\frac{299567}{2.50}$
 - Energy released by 1.00 g of methane = 119 827 J
- **Step 5:** Convert the answer from J to kJ:
 - $\frac{119827}{1000} = 120 \text{ kJ}$ (to 3 significant figures)



Your notes

 Exam Tip

- When new bonds are formed the amount of energy released is equal to the amount of energy absorbed when the same bonds are broken
- For example:



- Aqueous solutions of acid, alkalis and salts are assumed to be largely water so you can just use the m and c values of water when calculating the energy transferred
- To then calculate any changes in enthalpy per mole of a reactant or product the following relationship can be used:

$$\Delta H = -m \times c \times \Delta T$$

- When there is a rise in temperature, the value for ΔH becomes negative suggesting that the reaction is exothermic
- When there is a fall in temperature, the value for ΔH becomes positive suggesting that the reaction is endothermic
- Also, remember that ΔT is the same in $^{\circ}\text{C}$ and K !